UNIT V: Sustainable Manufacturing, Case Studies and Future Trends

Subject: 23A03703d - Smart Manufacturing

1. Sustainable and Green Manufacturing

1.1 Definition

Sustainable manufacturing creates products through processes that minimise negative environmental impact, conserve energy and natural resources, are safe for employees and communities, and are economically viable. Green manufacturing is the environmental part of this idea: less pollution, less waste, less energy and fewer harmful substances.

KEY: Sustainability has three pillars: environment (planet), economy (profit) and society (people). A good smart factory improves all three.

1.2 Why it matters

- Regulation: carbon taxes, emission limits and reporting rules (for example, the EU Carbon Border Adjustment Mechanism and corporate disclosure rules).
- Customers: buyers ask for low-carbon products and transparent supply chains.
- Costs: energy and material prices are volatile; efficiency protects margins.
- Risk: climate events and resource shortages disrupt production.

1.3 The 6 R's of green manufacturing

- Reduce: use fewer materials and less energy.
- Reuse: use parts or containers more than once.
- Recycle: turn scrap back into material.
- Recover: extract energy or materials from waste.
- Redesign: design products for disassembly and longer life.

- Remanufacture: restore used products to like-new condition.

1.4 Life cycle assessment (LCA)

LCA measures the environmental impact of a product from raw material to disposal. It counts energy, emissions, water and waste at every stage. It helps decide where improvement matters most.

- Cradle-to-gate: from raw material to the factory gate.
- Cradle-to-grave: the full life cycle including use and disposal.

2. Energy Efficiency in Smart Manufacturing

2.1 Where energy is used

In many plants, electric motors, compressed air, HVAC, lighting, furnaces and ovens consume most of the energy. Compressed air is often a surprisingly expensive utility because leaks waste large amounts of power.

2.2 Measures to improve efficiency

- Energy monitoring: sub-metering each machine line to find waste.
- Variable speed drives: match motor speed to load instead of running at full speed.
- Leak detection for compressed air: ultrasonic detectors find leaks quickly.
- Heat recovery: capture waste heat from furnaces and compressors.
- LED lighting with occupancy and daylight sensors.
- Power factor correction to reduce reactive losses and penalties.
- Load shifting: run energy-intensive processes when electricity is cheaper or greener.

2.3 Energy KPIs

- Energy per unit produced (kWh per part).
- Energy intensity (energy per unit of revenue or output).
- Carbon intensity (kg CO_2 per unit).
- Specific energy consumption for processes such as melting or drying.

KEY: You cannot improve what you do not measure: start with energy metering and a baseline.

2.4 Example: motor optimisation

A plant has 40 motors running at full speed. Adding variable speed drives to 15 fan and pump motors reduces their average consumption by 30 percent because power falls steeply with speed for fans and pumps. The payback period is 2 years, and the energy saving continues for the drive life.

2.5 Role of smart systems

- Digital energy management systems (EMS) collect meter data and show it on dashboards.
- Machine learning finds unexpected consumption and suggests setpoint changes.
- Digital twins test energy-saving changes before applying them.

Exam Q: Name three ways to reduce energy use in a compressed air system.

3. Sustainable Supply Chain Management

3.1 What it means

Sustainable supply chain management (SSCM) considers environmental and social impacts across all suppliers, transport and logistics, not only inside the factory. Many emissions and risks sit in the supply chain, often more than half of a product's footprint.

3.2 Key practices

- Supplier selection with sustainability criteria and audits.
- Low-carbon transport: optimised routes, consolidated loads, electric vehicles, rail where possible.
- Local sourcing where practical to cut transport emissions.
- Packaging reduction and returnable containers.
- Reverse logistics: taking back used products for reuse and recycling.

- Responsible sourcing of minerals and materials, with traceability.

3.3 Technology support

- IoT tracking of shipments and temperature for food and pharmaceuticals.
- Blockchain traceability for certifications and origin claims.
- AI route optimisation to reduce kilometres and fuel.
- Digital product passports that store material composition and recycling instructions.

3.4 Circular supply chains

A circular supply chain keeps materials in use for as long as possible. Products are designed for return, parts are remanufactured, and secondary materials replace virgin materials.

KEY: A circular supply chain needs data about the product: what it is made of, where it is, and when it comes back.

3.5 Example

A furniture maker tags each product with a QR code linking to a digital product passport. When a customer returns a sofa, the repair team knows the frame and fabric type, orders the right spare part, and refurbishes the item for resale.

Exam Q: Explain why supply chain emissions are important even for a factory with clean operations.

4. Role of AI in Green Manufacturing

4.1 AI for energy optimisation

- Predicts energy demand and suggests load schedules.
- Optimises setpoints for HVAC, compressors and furnaces.
- Detects unusual consumption that could mean leaks or faults.

4.2 AI for waste and material efficiency

- Vision systems detect defects early, reducing scrap and rework.

- Models optimise cutting patterns to reduce material offcuts.
- Process models find the best temperature and time settings for lower energy per part.

4.3 AI for product design and life cycle

- Generative design creates lighter parts with the same strength, using less material.
- AI estimates the environmental impact of design options during early design.
- Predictive maintenance extends equipment life and avoids wasteful breakdowns.

4.4 AI for supply chain sustainability

- Demand forecasting reduces overproduction and excess inventory.
- Route optimisation lowers transport emissions.
- Supplier risk models flag high-impact suppliers for audit.

4.5 Cautions

- AI systems use energy; the benefit must exceed the cost of computing.
- Models must be trained on good data and monitored for bias and drift.
- Human judgement remains important, especially for safety and ethics.

KEY: The best green AI project is one that saves more energy and material than it uses.

Exam Q: Give one example of how AI can reduce material waste in manufacturing.

5. Case Studies of Leading Smart Manufacturing Compani

5.1 Siemens: digitalised factories

Siemens operates digitalised factories, including the Amberg electronics plant in Germany. The company uses its digital twin software, PLC automation and MES to plan and operate lines. Its experience is often shared with customers as a reference model: integrated engineering, production and data.

- Lesson: connect engineering (design) and operations (production) to shorten change cycles.

5.2 Tesla: gigafactories and automation

Tesla's factories use large numbers of robots and an integrated approach to design and manufacturing, with many processes designed for high volume. Large castings and simplified designs reduce the number of parts. The company is known for rapid design changes in production.

- Lesson: design for manufacturing and automation from the start.
- Note: claims about production methods vary; verify with official reports before relying on specific figures.

5.3 Toyota: lean and smart production

Toyota's production system (lean manufacturing) is based on just-in-time delivery, jidoka (automation with a human touch: machines stop when a problem appears), and continuous improvement (kaizen). Smart technologies are being added to support these principles.

- Lesson: digital tools work best when they support sound process discipline.

5.4 Bosch: connected manufacturing

Bosch has built connected plants that use IoT sensors, edge computing and analytics to improve quality and reduce downtime. It also develops industrial IoT platforms for other manufacturers.

- Lesson: share knowledge between plants so that improvements scale quickly.

5.5 Comparing approaches

Company	Main focus	Key technology	Core lesson

Siemens	Integrated digital factory	Digital twin, PLC, MES	Link engineering and operations
Tesla	High-volume automation	Robots, design simplification	Design for manufacturing
Toyota	Lean production	Jidoka, JIT, kaizen	Process discipline first
Bosch	Connected plants	IoT, edge, analytics	Scale improvements across plants

Exam Q: Choose one case study and explain its lesson in your own words.

6. Emerging Technologies in Smart Manufacturing

6.1 5G and private wireless networks

5G offers low latency, high reliability and support for many devices. Private 5G networks inside a plant can connect mobile robots, AR devices and sensors with predictable performance.

6.2 Time-Sensitive Networking (TSN)

TSN adds deterministic timing to standard Ethernet. Control, vision and motion traffic can share one network without losing timing guarantees.

6.3 Generative AI and large language models

- Help engineers write PLC code, documentation and maintenance instructions.
- Let operators ask questions of machine manuals and historical records in plain language.
- Must be used with safeguards, because generated code and advice need verification before use on machines.

6.4 Autonomous systems and swarm robotics

Groups of mobile robots coordinate to move materials, inspect areas and assist in assembly without central control, improving flexibility and resilience.

6.5 Advanced materials and new manufacturing processes

- Metal additive manufacturing for lightweight and complex parts.
- Smart materials that change shape or report stress.
- Micro and nano manufacturing for electronics and medical devices.

6.6 Quantum computing (future outlook)

Quantum computers may in future solve certain optimisation problems, such as complex scheduling or materials simulation, much faster than classical computers. Practical industrial use is still at an early stage.

6.7 Edge AI and TinyML

Very small machine learning models run on low-power microcontrollers at sensors. This allows detection of vibration or sound patterns without any cloud connection.

KEY: Emerging technologies should be adopted according to a clear value case, not because they are new.

7. Future of Industry 5.0 and Beyond

7.1 What is Industry 5.0?

Industry 5.0 is a vision promoted by the European Union (around 2021) that complements Industry 4.0. It places the worker at the centre of production and adds sustainability and resilience as explicit goals.

- Human-centric: technology supports people, with cobots as partners rather than replacements; focus on skills, wellbeing and inclusion.
- Sustainable: energy and resource use are designed in from the start.
- Resilient: factories can keep operating during crises, such as supply shortages or disasters.

7.2 Differences between Industry 4.0 and 5.0

Aspect	Industry 4.0	Industry 5.0
Main focus	Efficiency and connectivity	Human values, sustainability and resilience
Role of people	Operators supervise machines	People and machines collaborate
Key technologies	IoT, cloud, AI, CPS	Adds human-machine interfaces, ethics and green tech
Goal	Productivity	Productivity with purpose

7.3 Skills for the future workforce

- Data literacy: understanding dashboards and models.
- Digital skills: programming, network and cybersecurity awareness.
- Systems thinking: seeing how processes, people and data interact.
- Adaptability and continuous learning.
- Problem solving and teamwork across engineering and business.

7.4 Future trends beyond Industry 5.0

- Fully autonomous, self-optimising factories with human supervision.
- Circular economy as the default business model.
- Digital twins of entire supply networks.
- Personalised products produced on demand at local micro-factories.
- Ethical AI and transparent decision systems on the shop floor.

7.5 Challenges ahead

- Cost and skills gap, especially for small and medium enterprises.
- Standards and interoperability across vendors.
- Cybersecurity as connectivity grows.
- Balancing automation with job quality.
- Climate targets that require deep reductions in energy and material use.

Exam Q: What does "human-centric" mean in Industry 5.0?

2.6 Carbon accounting: Scope 1, 2 and 3

Companies report greenhouse gas emissions in three scopes:

- Scope 1: direct emissions from sources the company owns, such as furnaces, boilers and company vehicles.
- Scope 2: indirect emissions from purchased electricity, heat and steam.
- Scope 3: all other indirect emissions across the value chain, such as purchased materials, transport, product use and end-of-life disposal.

Scope	Example in a factory	Who controls it
1	Gas burned in a furnace	The plant
2	Electricity bought from the grid	Energy supplier mix and plant efficiency
3	Steel purchased from a supplier	Suppliers and design choices

KEY: Scope 3 is often the largest share, which is why supply chain engagement is essential.

6.8 Green production technologies

- Cleaner production: changing the process to avoid pollution at source, for example replacing a toxic solvent with water-based cleaning.
- Water recycling: treating and reusing process water in cooling and washing, reducing intake and discharge.
- Zero liquid discharge (ZLD): systems that recover nearly all water from effluent so that only solids are removed.
- Dry machining and minimum quantity lubrication (MQL): less cutting fluid to dispose of and clean.
- Heat pumps and waste heat recovery for drying and heating processes.

- Renewable energy: on-site solar panels and power purchase agreements for wind or solar electricity.
- Bio-based and recycled materials for packaging and components.

5.6 Lessons from green case practice

- Start with a baseline: measure energy, water, waste and emissions before setting targets.
- Set specific targets such as reducing scrap by 15 percent or cutting energy per unit by 10 percent in two years.
- Engage employees: ideas from operators often produce the best low-cost improvements.
- Report transparently using recognised standards such as GRI or CDP.
- Link sustainability KPIs to the same dashboards used for productivity, so they are managed together.

7.6 Ethics and human-centric design

- Transparency: workers should know what data is collected about them and how it is used.
- Fairness: AI decisions on scheduling or performance must be checked for bias.
- Safety: cobots and autonomous vehicles must be designed around human behaviour and error.
- Inclusion: interfaces should suit workers of different ages, languages and abilities.
- Participation: workers should help design new workstations and processes.

- Job transition: retraining and redeployment programmes should accompany automation projects.

KEY: Industry 5.0 asks not only "can we automate this?" but also "should we, and how will it affect people?"

7.7 Resilience in practice

- Dual sourcing of critical parts to avoid single-supplier failure.
- Digital twins of supply networks to test disruption scenarios.
- Buffer stock for high-risk items and flexible production lines that can switch products.
- Energy backup and microgrids to keep critical loads running during outages.
- Regular risk drills that combine IT, operations and procurement.

8. Worked Examples and Revision

8.1 Worked example: green business case

A plant wants to cut its compressed air cost. Current cost: 120,000 per year. Survey finds 25 percent leakage. Fixing leaks costs 15,000 and saves about 25 percent of the cost, which is 30,000 per year.

- Simple payback = 15,000 / 30,000 = 0.5 year (six months).
- Add a variable speed compressor later for further savings.
- Result: a quick, strong business case that also reduces emissions.

8.2 Worked example: circular packaging

A company replaces single-use cardboard boxes with a returnable plastic crate used 40 times. If each crate replaces 40 boxes and each box costs 1.5 to produce and ship, the savings per crate cycle are visible in the cost and the waste reduction. Tracking each crate with RFID shows where crates are lost and how to reduce loss.

8.3 Revision checklist

- Can you define sustainable and green manufacturing?
- Can you list the 6 R's?
- Can you explain life cycle assessment in one sentence?
- Can you give three ways to reduce energy in a plant?
- Can you describe one AI application for green manufacturing?
- Can you summarise one case study and its lesson?
- Can you compare Industry 4.0 and Industry 5.0?

8.4 Short answer practice

Exam Q: Why is supply chain sustainability important for a manufacturer?

Exam Q: What is the difference between recycling and remanufacturing?

Exam Q: Name two emerging technologies and one possible use for each in a factory.